

Supplementary information

Spatial association of TLS programs and LAMP3⁺CCR7⁺ mregDC states marks an immune-prognostic context in ovarian cancer

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Supplementary Figure Legends

Supplementary Figure S1. singleCellHaystack spatial non-randomness audit.

A, Top spatially non-random genes ranked by singleCellHaystack evidence in the deterministic ovarian cancer subset. **B**, Marker-level audit for TLS/mregDC/AP-associated genes, grouped by AP/MHC-II proxy, mregDC/AP, and TLS lymphoid marker categories.

Supplementary Figure S2. Hotspot spatial-autocorrelation audit.

A, Top spatially autocorrelated genes ranked by Hotspot Z score in sampled Xenium cells. **B**, Marker-level Hotspot audit for TLS/mregDC/AP-associated genes, grouped by AP/MHC-II proxy, mregDC/AP, and TLS lymphoid marker categories.

Supplementary Figure S3. Xenium spatial maps supporting feature audits.

A, Spatial module maps for TLS-integrated, mregDC/TLS-axis, mregDC-extended, AP/MHC proxy, and IF-core proxy scores. **B**, Representative spatial expression maps for singleCellHaystack-prioritized non-random genes. **C**, TLS-score map and TLS/mregDC co-high cells in the same Xenium coordinate framework. **D**, Representative spatial expression maps for Hotspot-prioritized autocorrelated genes.

Supplementary Figure S4. Expanded multi-sample spatial source-data validation.

A, Spot-level co-high enrichment across public ovarian spatial transcriptomic samples. **B**, Gene-set coverage audit for TLS, TLS-imprint, mregDC, MHC-II, IF-core, and T-cell-related signatures. **C**, Exact-marker availability for HLA-DRA, LAMP3, and CCR7 across the same datasets.

Supplementary Figure S5. Exploratory model validation.

A, Time-dependent ROC curves for 1-, 3-, and 5-year internal model assessment in TCGA-OV. **B**, LASSO-Cox cross-validation curve with the selected penalty marker. **C**, Calibration-by-risk-quintile summaries for 1-, 3-, and 5-year predicted survival.

Supplementary Figure S6. Immune-richness-adjusted specificity audit.

A, Raw and immune/B/T-cell/tumor-purity-adjusted Spearman correlations for mregDC-TLS, mregDC-AP/MHC-II, and TLS-AP/MHC-II associations in TCGA-OV patients. **B**, Raw and local B/T/AP-adjusted correlations for the same relationships across spatial transcriptomic spots. **C**, Raw and AP/IFN/tumor-adjusted correlations in Xenium cells.

Supplementary Figure S7. Exploratory drug-prioritization and docking-audit summary.

A, Compound-by-evidence matrix showing standardized support across clinical/target evidence, virtual screening, predicted risk/TLS/axis/IF/AP program reversal layers, and docking support. **B**, Integrated candidate priority, computed as the mean standardized value across the displayed evidence layers. **C**, Relationship between integrated priority and docking support for compounds with target-class color coding; selected compounds are labeled to show concordant and discordant prioritization patterns. **D**, Target-class-level summary of target evidence, program-reversal support, and docking support.

This figure is an exploratory computational prioritization layer only. It does not use matched IC50 or PIK3CA mutation data, does not establish drug efficacy, and is not used as primary evidence for the TLS/mregDC spatial conclusion.

Supplementary Source Data

- Current-numbering source-data map for main and supplementary figures:
source_data/SOURCE_DATA_INDEX.csv.
- Fig. S1: source_data/FigS01_singleCellHaystack_audit.
- Fig. S2: source_data/FigS02_Hotspot_audit.
- Fig. S3: source_data/FigS03_Xenium_spatial_maps.
- Fig. S4: source_data/FigS04_multi_sample_spatial_source.
- Fig. S5: source_data/FigS05_model_validation.
- Fig. S6: source_data/FigS06_specificity_adjustment.
- Fig. S7: source_data/FigS07_drug_prioritization_docking.
- Signature definitions, GEO source reconstruction, public spatial and 10x dataset

accessions, and gene-coverage audits are provided in Supplementary Tables S1-S3 and `source_data/Tables_signature_and_GEO`.

- Legacy source-data folders are retained for traceability but should be interpreted through the current-numbering index.

Supplementary Tables

Supplementary Table S1.

Reconstructed 22-sample scRNA-seq GEO source manifest, disease labels, sample counts, and accession-level summaries.

Supplementary Table S2.

Signature definitions, gene-coverage audits, marker-availability checks, and legacy-label harmonization for the mregDC, TLS, antigen-presentation/MHC-II, IFN, FDC/GC, HEV/stromal, proliferation, and EMT/invasion modules.

Supplementary Table S3.

Twenty-three public multi-sample spatial transcriptomic samples used for Fig. 5/Fig. S4, including one public 10x Visium ovarian FFPE sample within the 23-sample spatial set, dataset-level GSE identifiers, sample-level GSM identifiers where applicable, and spot/cell counts. The same table also lists the separate public 10x scFFPE and Xenium Prime ovarian cancer reference datasets with URLs and software-version notes.